

Force

In physics, a **force** is a push or pull acting upon an object as a result of the object's interaction with another object.

It is an external agent capable of altering an object's state of rest or motion, causing it to accelerate, decelerate, or change direction.

Key Characteristics

- **Vector quantity** – Forces have both magnitude (strength) and direction.

- **SI unit** – The standard metric unit for force is the **Newton (N)**.

One Newton is the force required to accelerate a 1 kg mass at a rate of 1 m/s².

- **Formula** – Newton's Second Law of Motion:

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$$\mathbf{F} = m\mathbf{a}$$

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where **F** is force, **m** is mass, and **a** is acceleration.

Common Types of Forces

Contact Forces

Applied through physical contact between objects.

Examples:

- Friction

- Tension

- Air resistance

- Physical push/pull

Non-Contact Forces

Act over a distance without physical contact.

Examples:

- Gravitational
- Electromagnetic
- Strong nuclear
- Weak nuclear

Summary

- A force changes the motion of an object.
- It is a vector quantity measured in Newtons.
- Newton's Second Law links force to mass and acceleration.
- Forces are broadly divided into contact and non-contact types, each with familiar examples.
